

FormulaBench

Auditable Excel formula generation

Research / Track 2

Encode x Ylookup Rebuild Private Markets Hackathon

```
=IFERROR(  
    INDEX>Returns!$H:$H,  
  
    MATCH($B9>Returns!$B:$B,0)),  
    0  
)
```

INSPECTED

GENERATED

VALIDATED

Plausible formulas fail in four different ways

A workbook can open successfully while still producing the wrong investment result.

01

Context

The relevant worksheet, source range or existing formula never reaches the model.

02

Coverage

A response omits required cells, assigns one twice or writes outside the target.

03

Workbook

A formula introduces an unsafe reference, invalid type or broken workbook package.

04

Semantics

The formula executes, yet its logic or transformation still produces the wrong value.

Execution and correctness require separate checks.

FormulaBench system

One model response sits inside a deterministic workbook pipeline.



MODEL BOUNDARY

Temperature 0 · thinking disabled · one response · no unrestricted tools

DETERMINISTIC BOUNDARY

Exact target coverage and workbook checks run before any proposed edit reaches the output.

The output contract blocks silent corruption

Every accepted response must satisfy the same workbook-level contract.

Check	Required evidence	Failure behaviour
Sheet + A1	Every target carries an explicit worksheet and cell address.	Reject the full response
Exact coverage	Finite targets contain no gaps, duplicates, overlaps or extra assignments.	Reject the full response
Typed values	Dates and datetimes use explicit Excel value types rather than locale-dependent text.	Reject invalid schema
Formula gate	External paths, URIs, DDE and high-risk executable or data functions remain blocked.	Reject unsafe formula
Atomic reopen	The writer saves a fresh copy, reopens it and verifies the intended cell state.	Publish pristine fallback

Rejected responses leave the supplied workbook unchanged.

Public self-evaluation: 133 of 400 tasks passed

Complete public benchmark after LibreOffice recalculation using the organiser-supplied evaluator.

33.25%

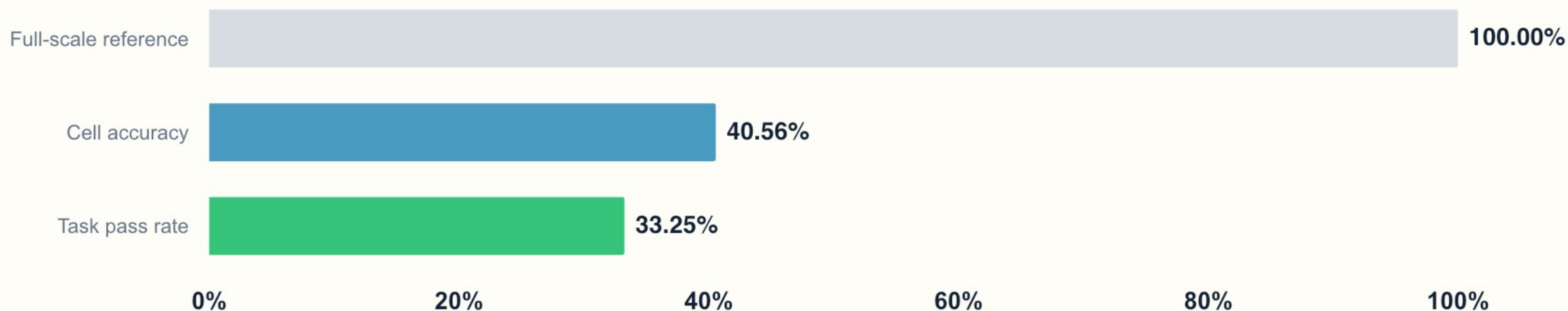
task pass rate

40.56%

cell accuracy

400/400

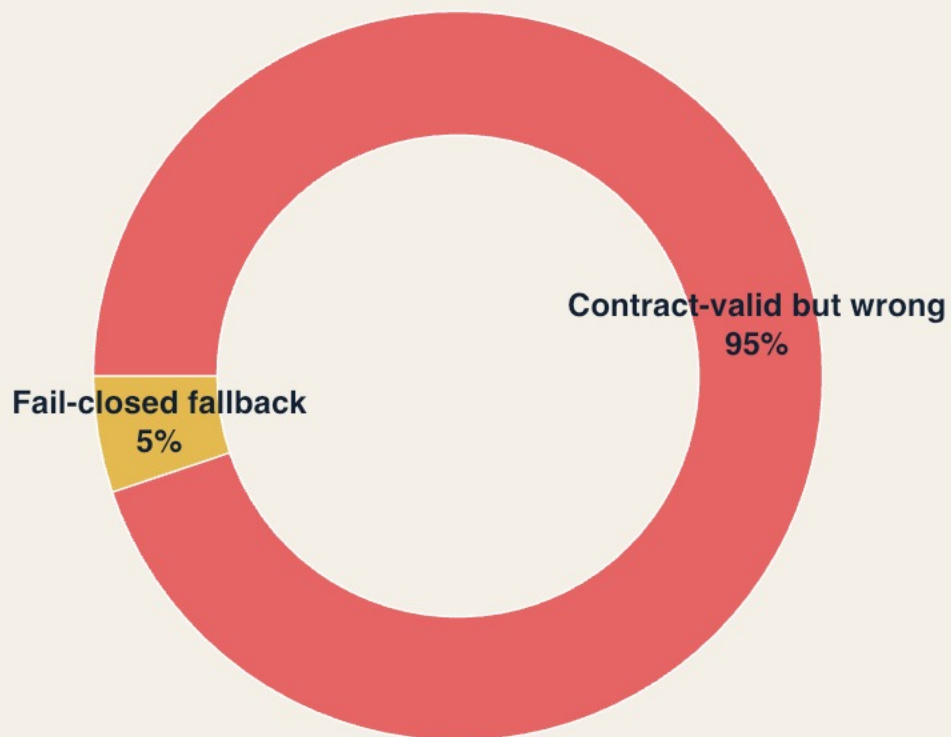
evaluated · 0 missing · 0 errors



120,824 of 297,882 target cells matched the golden workbook.

Most wrong cells came from accepted workbooks

Strict rejection explains 5.16% of wrong cells. Semantic errors explain the rest.



94.84%

of wrong cells sit inside 221 responses that passed the structural contract.

84.76%

of all wrong cells came from the four transformations with more than 10,000 target cells.

Context truncated: 24.10% passed

Context not truncated: 41.95% passed

This association does not establish causation.

The organiser reference and FormulaBench test different systems

The reported pass rates should not support a controlled causal comparison.

≈59%

organiser reference

33.25%

FormulaBench public self-evaluation

Same base model

Different inference and output contracts

System	Inference mode	Requested output	Validation evidence
Organiser reference	xhigh reasoning inferred from the published runner and cookbook	Final scalar values	≈59% reported. Run traces and renderer metadata not published
FormulaBench	Thinking disabled, temperature 0, one native Tinker sample	Executable formulas and typed transformations	Organiser-supplied evaluator with exact coverage checks and full traces

Direct attribution of the gap to one component would be misleading.

Next version: choose the operation before producing cells

The failure analysis points to two execution paths.

FORMULA TASKS

Synthesize a formula

Recalculate locally and test explicit invariants before publishing the workbook.

LARGE TRANSFORMATIONS

Select an operation

Let a deterministic executor handle filter, sort, append, group, deduplicate and clear operations.

LIVE RESEARCH DEMO

masterasnackin.github.io/FormulaBench/

PUBLIC REPOSITORY

github.com/MasteraSnackin/FormulaBench

The organiser private benchmark remains the genuine unseen test.